

CHAPTER 4 SYSTEM DESIGN

4.1 System Architecture

The system design of **Mentrix – AI-Powered Student Academic Assistant** describes how the different parts of the system work together to provide document-based academic assistance. The design focuses on document processing, AI interaction, personalized learning features, data management, and user interaction.

4.1.1 Overall System Working

The overall working of Mentrix is divided into the following stages:

1. User Authentication

- a. The student registers or logs into the Mentrix application.
- b. JWT is used to authenticate the student.

2. Document Upload

- a. The student uploads academic material such as PDF, DOCX, PPT/PPTX files, and images.
- b. The uploaded files are stored in MinIO.

3. Document Processing

- a. The uploaded documents are processed by the AI service.
- b. Text is extracted from supported documents.
- c. OCR is performed on scanned documents and images.
- d. The extracted content is divided into smaller chunks.

4. Knowledge Storage

- a. Embeddings are generated from the processed document chunks.
- b. The embeddings are stored in Qdrant for semantic search.

5. AI Interaction

- a. Students can ask questions about their uploaded documents.
- b. The RAG pipeline retrieves relevant information from Qdrant.
- c. Gemini generates an answer using the retrieved content.

6. Personalized Learning

- a. Quizzes are generated from academic content.
- b. Quiz performance is analysed to identify weak topics.

- c. Flashcards are generated according to the identified weak areas.
- d. AI-generated notes are created from selected documents or topics.

7. Digital Note-Taking

- a. Students can use the whiteboard for simple notes, drawings, and diagrams.

4.1.2 Technology Used

Component	Technology	Purpose
Frontend	React.js, TypeScript	User interface
State Management	Redux Toolkit	Application state
Backend	Node.js, Express.js	APIs and business logic
Authentication	JWT, bcrypt	Authentication and password security
AI Service	Python, FastAPI	AI and document processing
AI Framework	LangChain	RAG and AI workflows
AI Model	Google Gemini API	AI generation and OCR
Database	PostgreSQL	Structured application data
ORM	Prisma	Database operations
Vector Database	Qdrant	Embedding storage and semantic search
File Storage	MinIO	Academic document storage
Cache/Queue	Redis	Caching and background processing
Whiteboard	Excalidraw API	Digital note-taking
Deployment	Docker, Docker Compose	Containerized deployment

4.2 System Interface Design

The system interface of **Mentrix** is designed to provide students with a simple and organized environment for accessing academic resources and AI-powered learning features. The interface follows a consistent layout with a navigation sidebar, workspace area, document selection, action buttons, status indicators, and form controls. The major interfaces of the system are described below.

4.2.1 Login Interface

The Login Interface provides authenticated access to the Mentrix system.

Main elements:

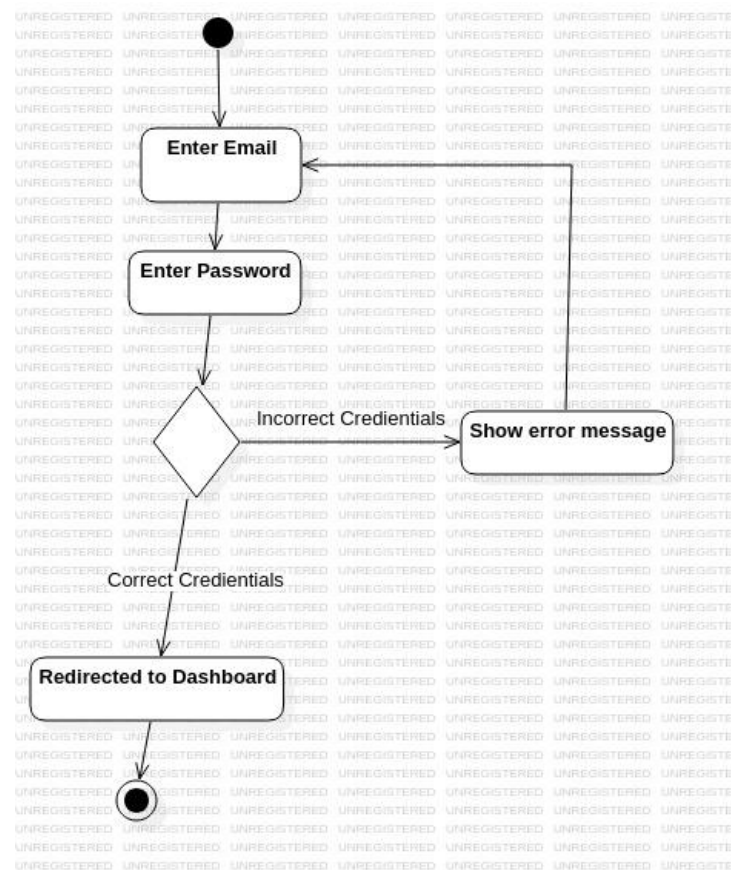
- Email address field for entering the registered email.
- Password field for entering the account password.
- Password visibility option.
- **Sign In** button for authentication.
- **Create Account** option for new users.

Validation:

- Email address must be provided in a valid format.
- Password must be entered.
- Invalid credentials are rejected.
- Only authenticated users can access the main application.

Steps:

1. Enter the registered email address.
2. Enter the password.
3. Click **Sign In**.
4. The system validates the credentials.
5. On successful authentication, the user is redirected to the dashboard.



Flowchart 4.2.1: Login

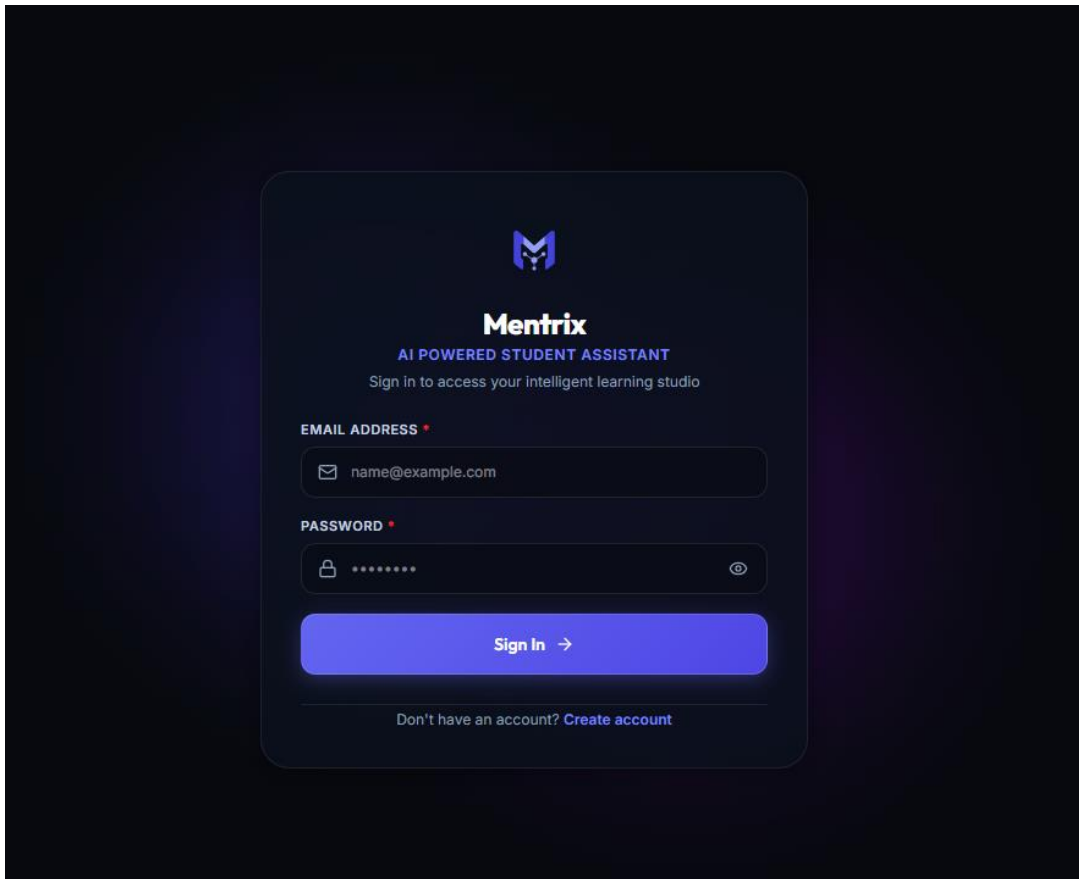


Figure 4.2.1: Login Interface

4.2.2 User Registration Interface

The Registration Interface allows a new student to create a Mentrix account.

Main elements:

- Username field.
- Email address field.
- Password field.
- Gender selection.
- Age field.
- Optional profile picture.
- **Complete Registration** button.

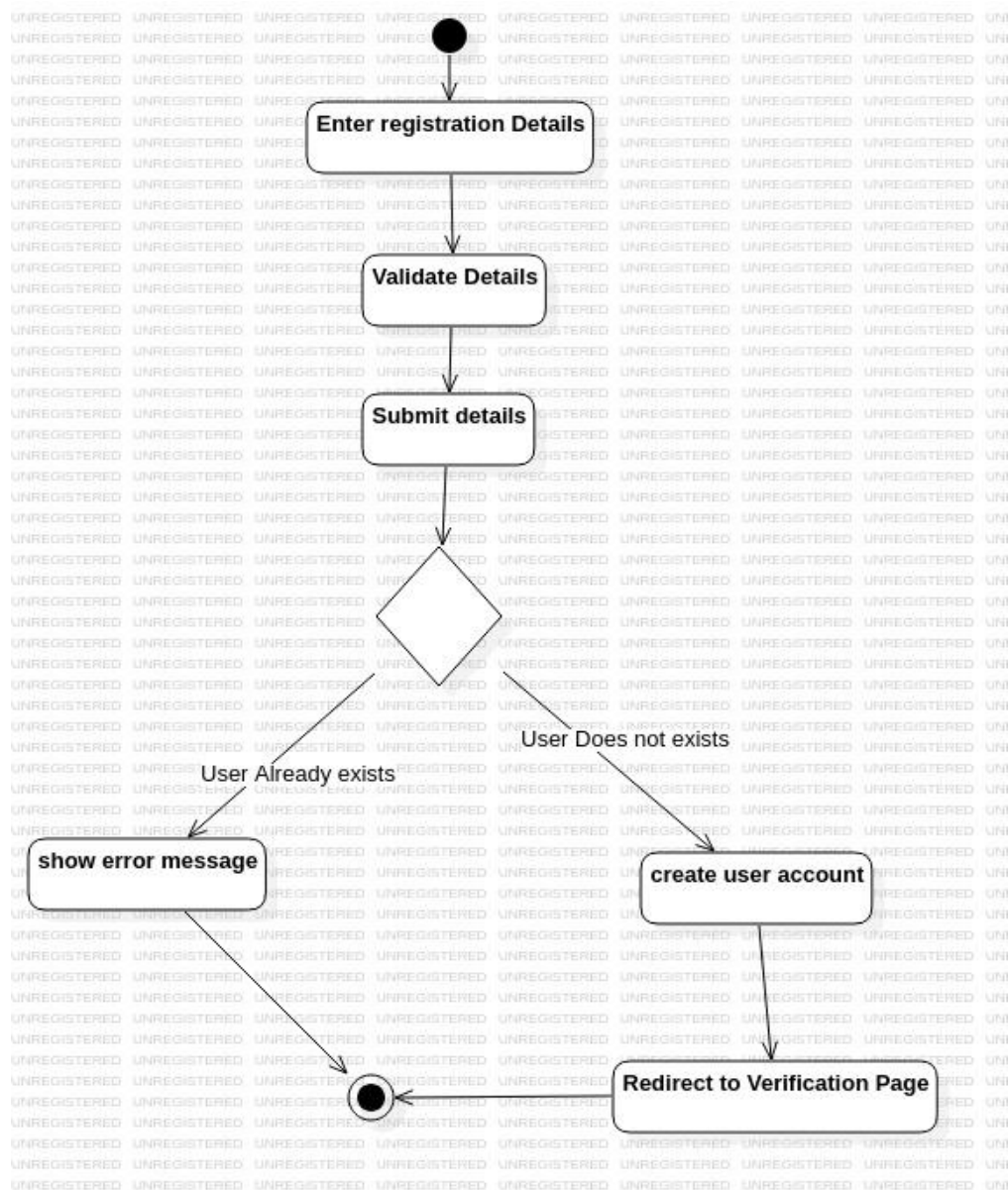
Validation:

- Username, email, password, gender and age are required.
- Email must be in a valid format.

- Password must satisfy the minimum length requirement.
- Age must contain a valid numerical value.
- Existing email addresses cannot be registered again.

Steps:

1. Enter the username.
2. Enter the email address.
3. Create a password.
4. Select gender and enter age.
5. Optionally add a profile picture.
6. Click **Complete Registration**.
7. The system processes the registration and proceeds to email verification.



Flowchart 4.2.2: Registration

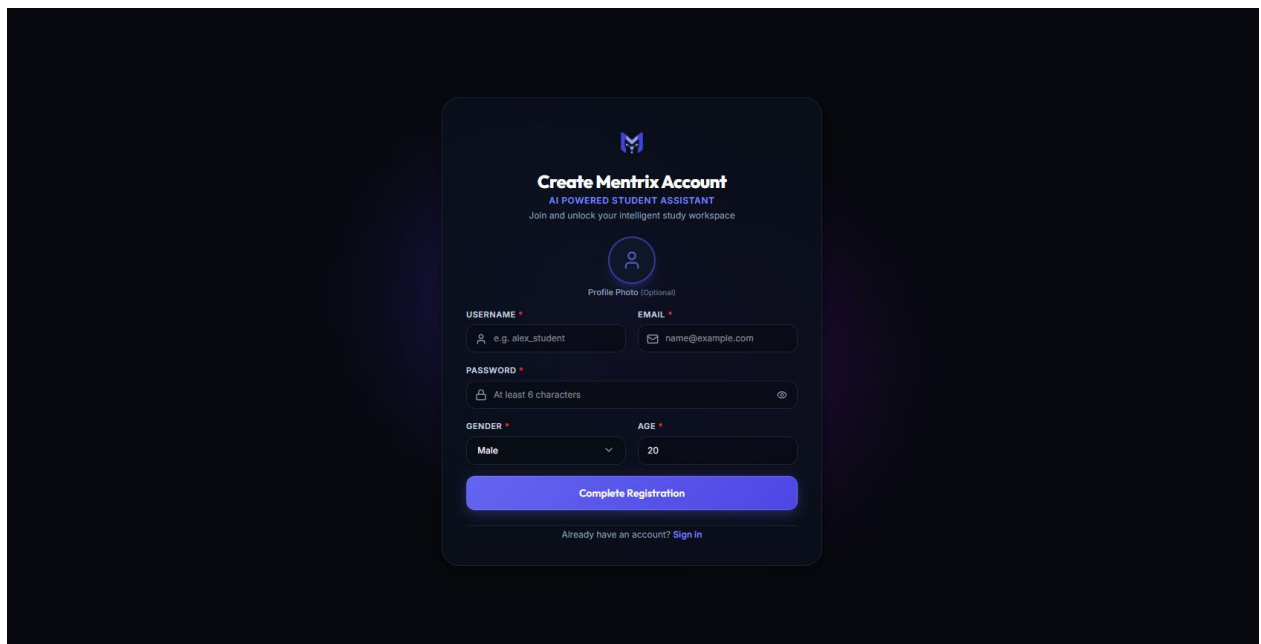


Figure 4.2.2: Registration Interface

4.2.3 Email Verification Interface

The Email Verification Interface verifies the student's email address using a six-digit verification code.

Main elements:

- Six-digit OTP input fields.
- **Verify & Proceed** button.
- **Resend** option.
- Back to Registration option.
- Sign In option.

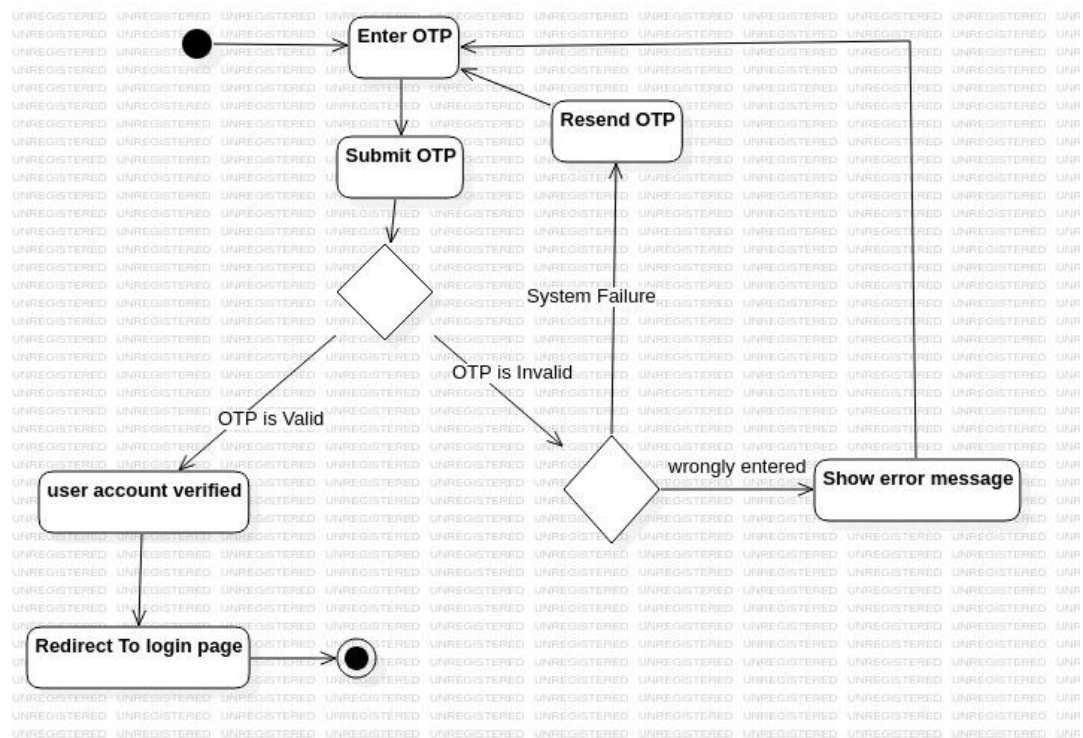
Validation:

- A six-digit verification code must be entered.
- The entered code must match the code sent to the registered email.
- Expired or incorrect codes are rejected.
- Resending the code is available after the specified waiting period.

Steps:

1. Register a new Mentrix account.
2. Receive the verification code through email.

3. Enter the six-digit code.
4. Click **Verify & Proceed**.
5. The system validates the code.
6. After successful verification, the account becomes verified.



Flowchart 4.2.3: Email Verification

The screenshot shows a mobile app interface for email verification. At the top is a blue 'M' logo. Below it, the title 'Verify Your Email' is displayed in bold. Underneath the title, it says 'Enter the 6-digit verification code sent to' followed by an email address 'ironman@mail.com' in a blue box. Below this is a row of six empty rounded square boxes for entering the code, with the text '0 of 6 digits entered' below them. A large blue button with a checkmark icon and the text 'Verify & Proceed →' is positioned below the input boxes. At the bottom, there is a link 'Didn't receive the code? Resend in 58s' and two smaller links: '← Back to Register' and 'Sign In Instead'.

Figure 4.2.3: Email Verification Interface

4.2.4 Dashboard Interface

The Dashboard acts as the main workspace of Mentrix. It provides an overview of the student's learning activities and quick access to the major features.

Main elements:

- Student profile information.
- AI engine status.
- Daily learning goal.
- Learning streak.
- Document count.
- Practice quiz count.
- Study notes count.
- Whiteboard count.
- Quick actions for learning features.
- Navigation menu for different modules.
- Document upload option.

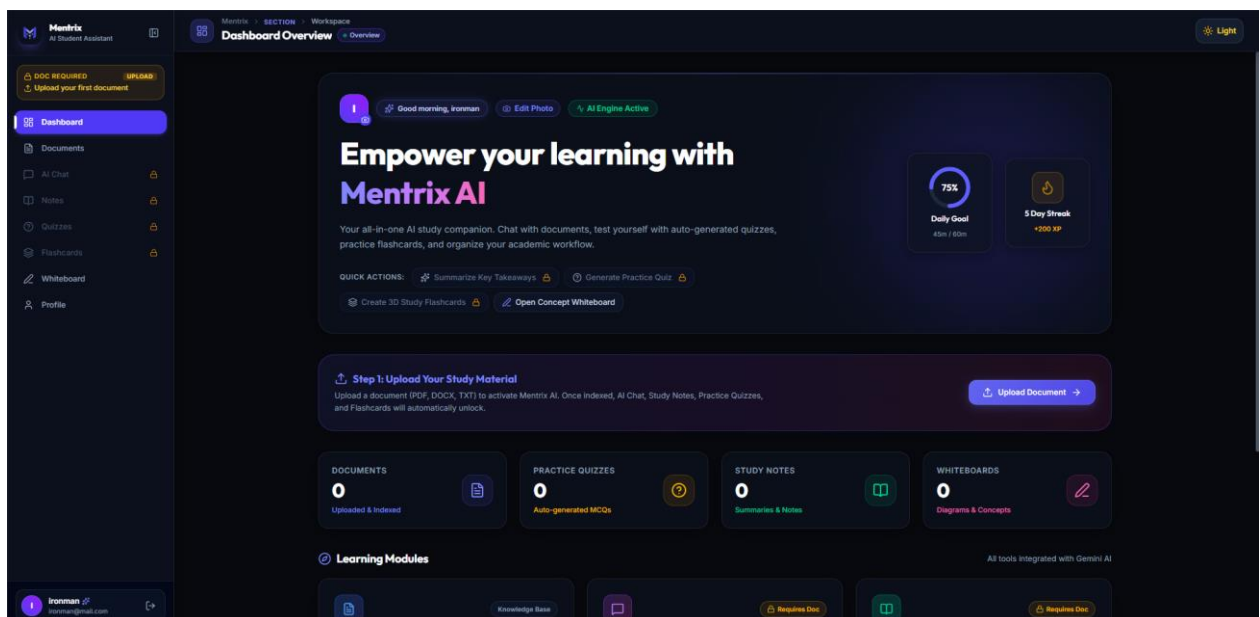


Figure 4.3.4: Dashboard Interface

4.2.5 Document Upload Interface

The Document Interface allows students to upload their academic study material. The uploaded material acts as the source for AI-based learning features.

Supported study material shown in the interface:

- PDF
- DOCX
- TXT

Main elements:

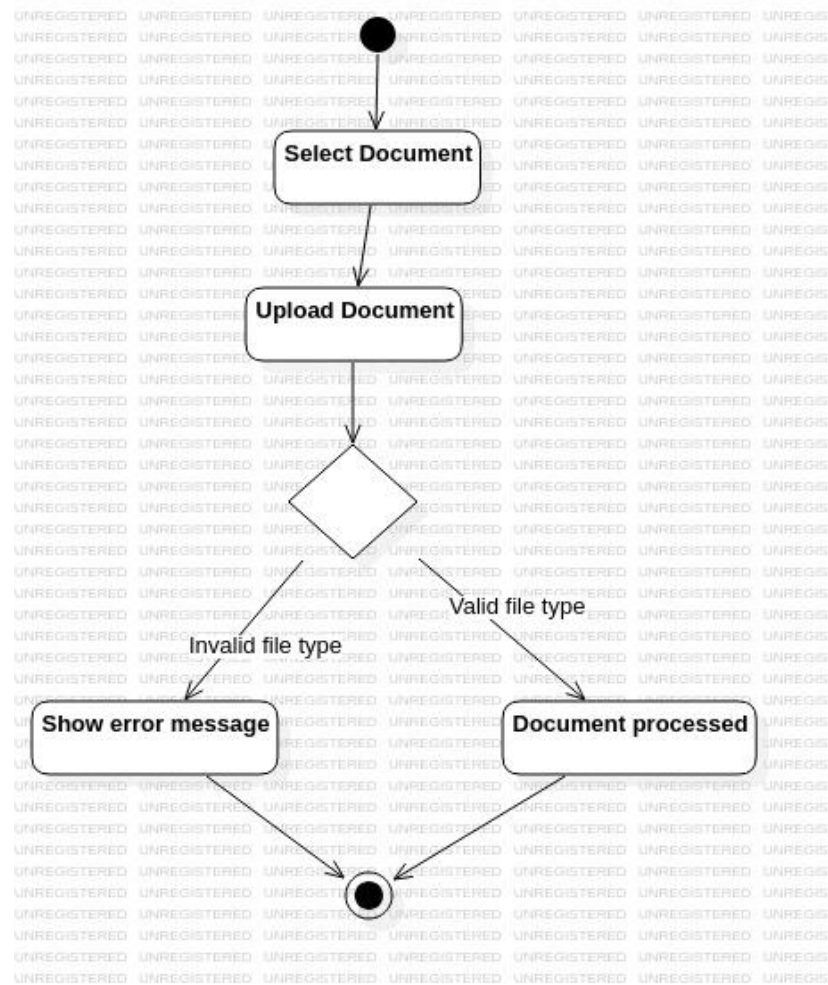
- Drag-and-drop upload area.
- File selection option.
- Upload status.
- List of uploaded documents.
- Active document selection.

Validation:

- Only supported file formats are accepted.
- The file size must remain within the specified limit.
- Invalid file types are rejected.
- Uploaded files are processed before becoming available to AI features.

Steps:

1. Open the **Documents** section.
2. Select a file or drag and drop it into the upload area.
3. The system validates the selected file.
4. The file is uploaded and processed.
5. The document becomes available for AI-based features after processing.



Flowchart 4.2.5: Document Upload

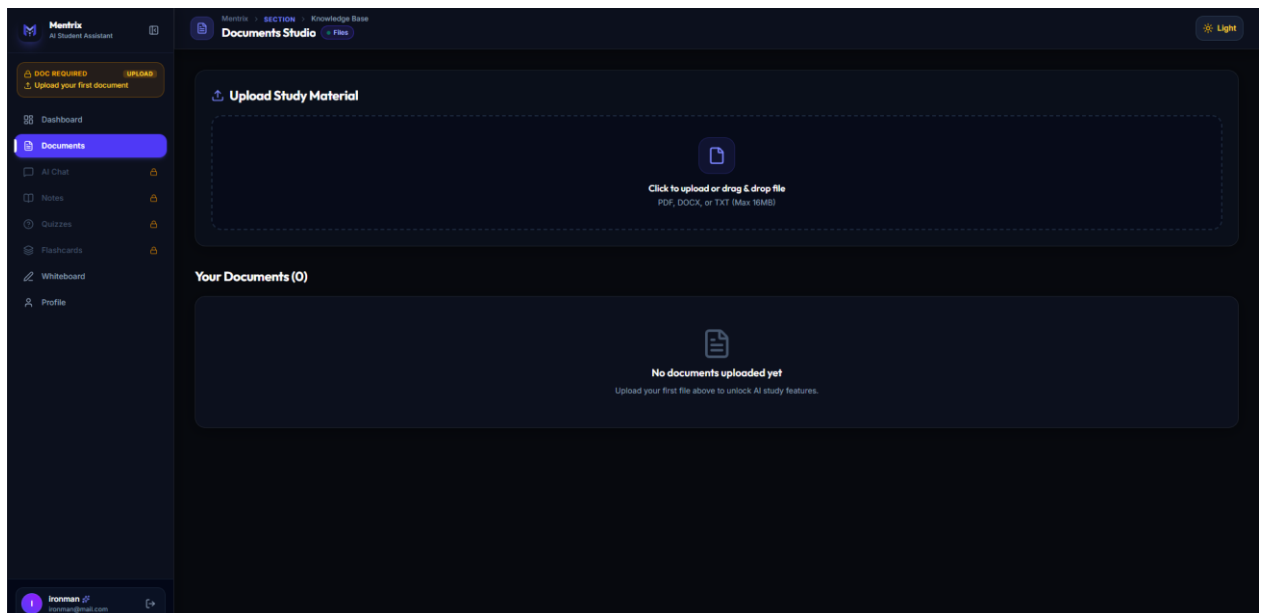


Figure 4.2.5: Document Upload Interface

4.2.6 AI Chat Interface

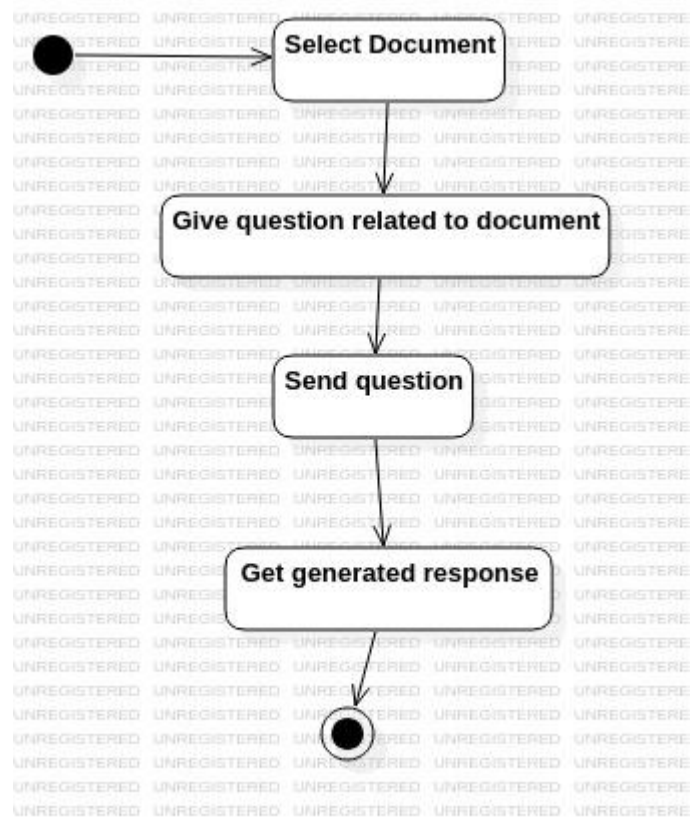
The AI Chat Interface allows students to ask questions about the selected academic document. The interface displays the active document and provides a conversational workspace.

Main elements:

- Active document indicator.
- AI assistant status.
- Chat/message area.
- Suggested questions.
- Message input field.
- Send button.

Steps:

1. Upload and select an academic document.
2. Open **AI Chat**.
3. Select the required document as the context.
4. Enter a question related to the document.
5. Click the send button.
6. The AI processes the question and provides a response based on the available document content.



Flowchart 4.2.6: AI Chart

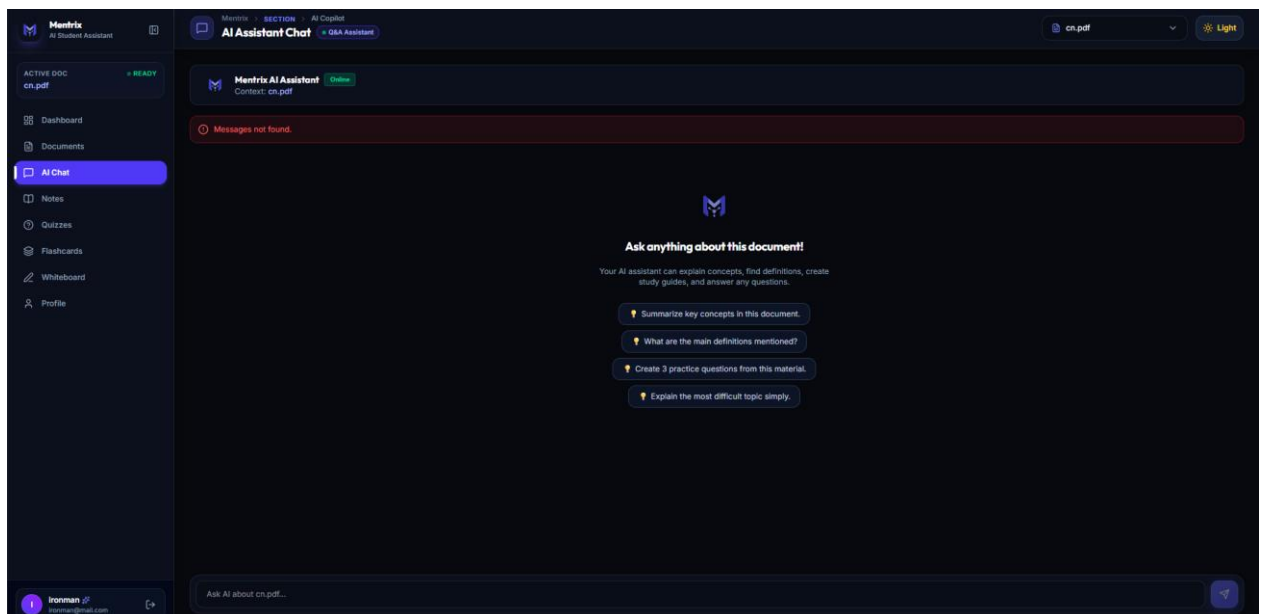


Figure 4.2.6: AI Chat Interface

4.2.7 Quiz Interface

The Quiz Interface allows students to generate and attempt AI-based practice quizzes from their study material.

Main elements:

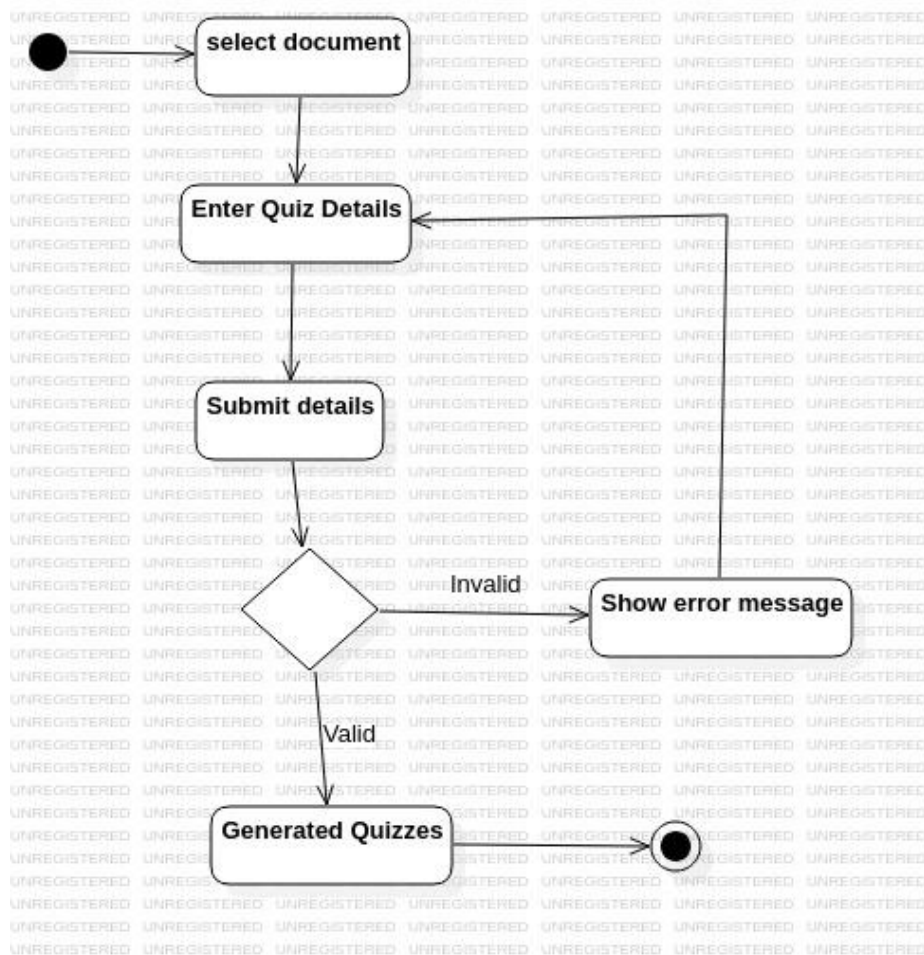
- Quiz title field.
- Difficulty selection.
- Number of questions selection.
- **Create Quiz** button.
- Quiz Library.
- Quiz practice area.

Validation:

- Quiz title must be provided.
- A valid difficulty level must be selected.
- Number of questions must be selected.
- A document must be available for generating document-based quizzes.

Steps:

1. Select an academic document.
2. Open the **Quizzes** section.
3. Enter a quiz title.
4. Select the difficulty level.
5. Select the required number of questions.
6. Click **Create Quiz**.
7. The AI generates the practice quiz.
8. Select the generated quiz from the Quiz Library.
9. Attempt the questions and submit the quiz.



Flowchart 4.2.7: Quizzes

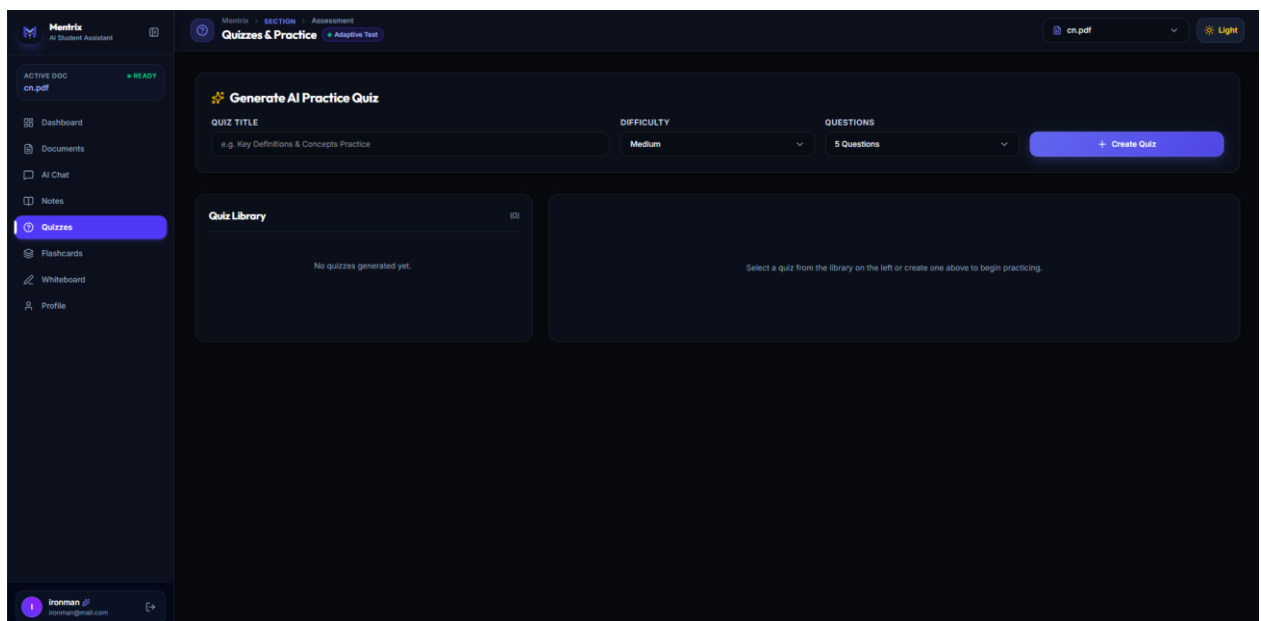


Figure 4.2.7: Quiz Interface

4.2.8 Flashcard Interface

The Flashcard Interface provides AI-generated flashcards for revision. Flashcards are connected with quiz performance so that students can use them to improve their understanding of weak areas.

Main elements:

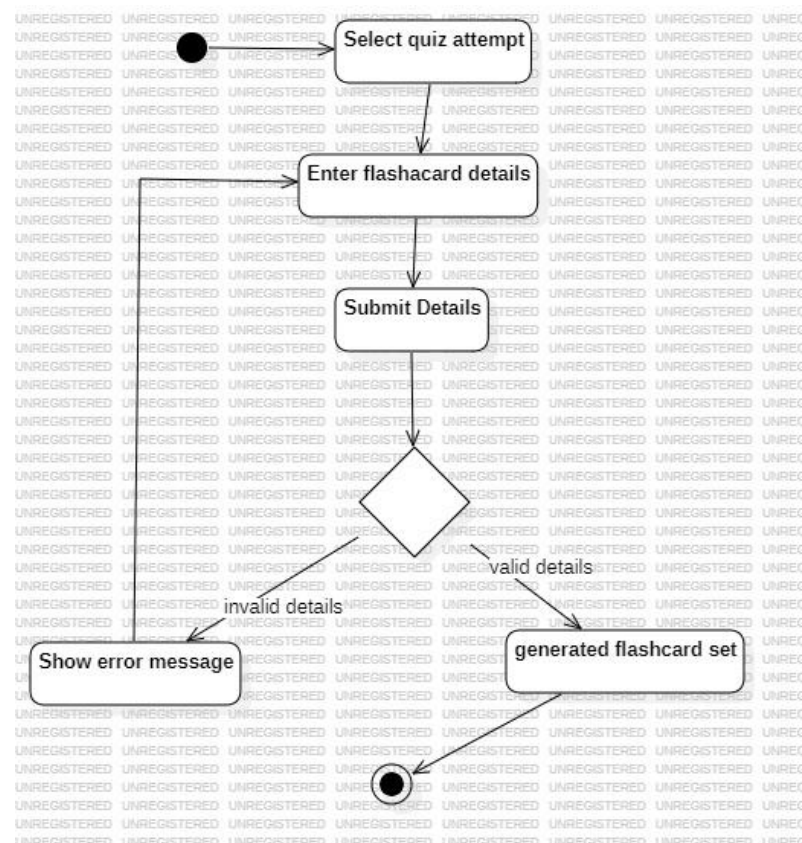
- Selected document indicator.
- Flashcard library.
- Quiz attempt information.
- Generated flashcard sets.
- Flashcard study area.

Validation:

- A quiz attempt must be available before flashcards can be generated.
- Flashcards are generated from quiz-related learning content.

Steps:

1. Select a quiz attempt.
2. Enter flashcard details such as difficulty, no. Of cards and title
3. Click create
4. AI generates relevant flashcards.
5. The student reviews the generated flashcards for revision.



Flowchart 4.2.8: Flashcard Generation

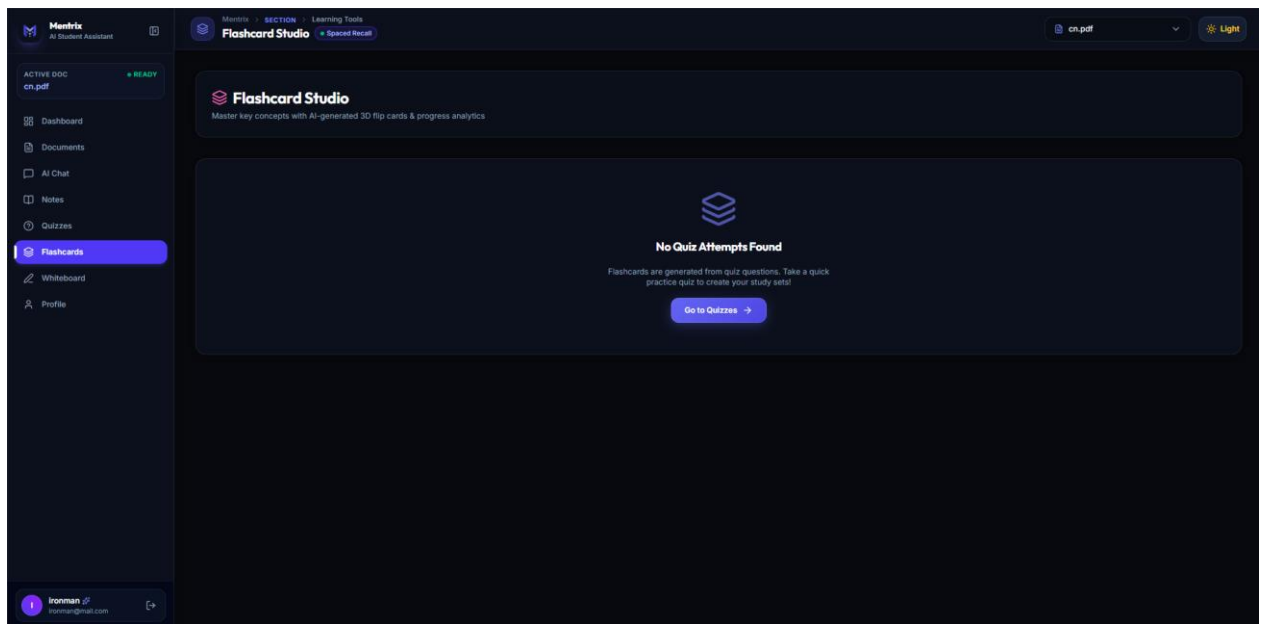


Figure 4.2.8: Flashcard Interface

4.2.9 AI Notes Interface

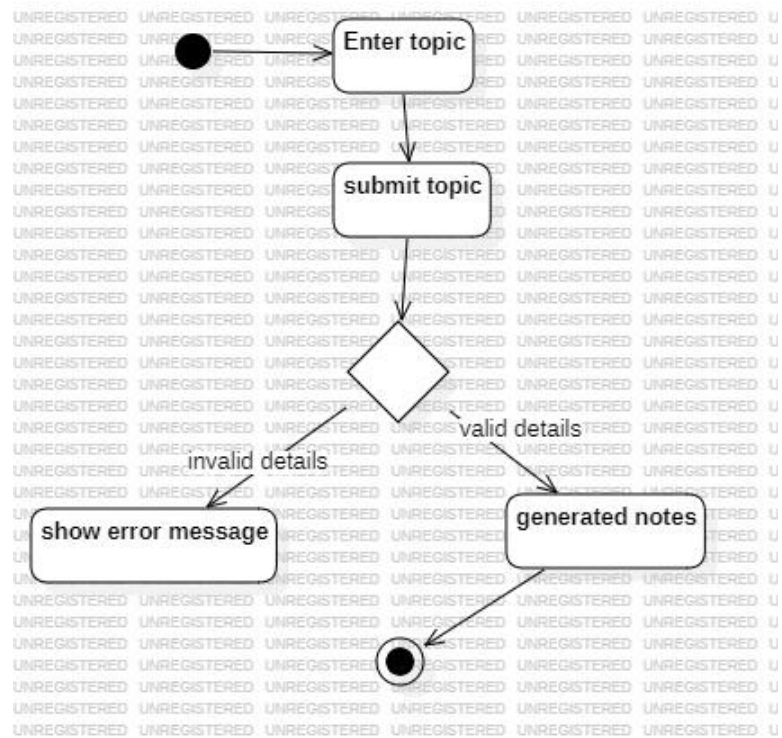
The AI Notes Interface allows students to create and manage study notes. Notes can contain manually written content as well as AI-generated learning insights.

Main elements:

- Notes Library.
- **New Note** button.
- Note title field.
- Note content editor.
- Write/Preview options.
- **Save Note** button.
- Word and character counter.

Steps:

1. Open the **Notes** section.
2. Click **New Note**.
3. Enter the note title.
4. Add or generate the required study content.
5. Review the content using the Preview option.
6. Click **Save Note**.
7. The note is stored in the Notes Library.



Flowchart 4.2.9: AI Notes

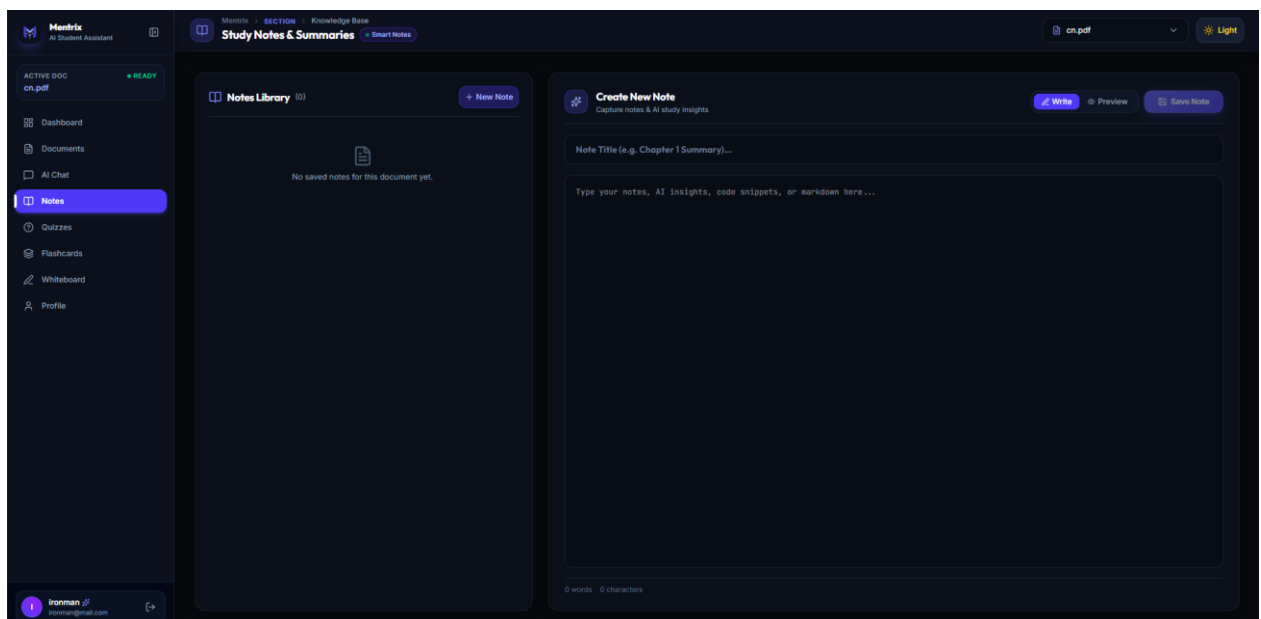


Figure 4.2.9: AI Notes Interface

4.2.10 Whiteboard Interface

The Whiteboard Interface provides students with a digital workspace for making simple handwritten notes, diagrams, and visual representations.

Main elements:

- Drawing canvas.

- Draw tool.
- Eraser tool.
- Pen colour selection.
- Pen size control.
- Clear option.
- Export option.
- **New Canvas** button.
- **Save to Cloud** button.
- Saved Boards section.

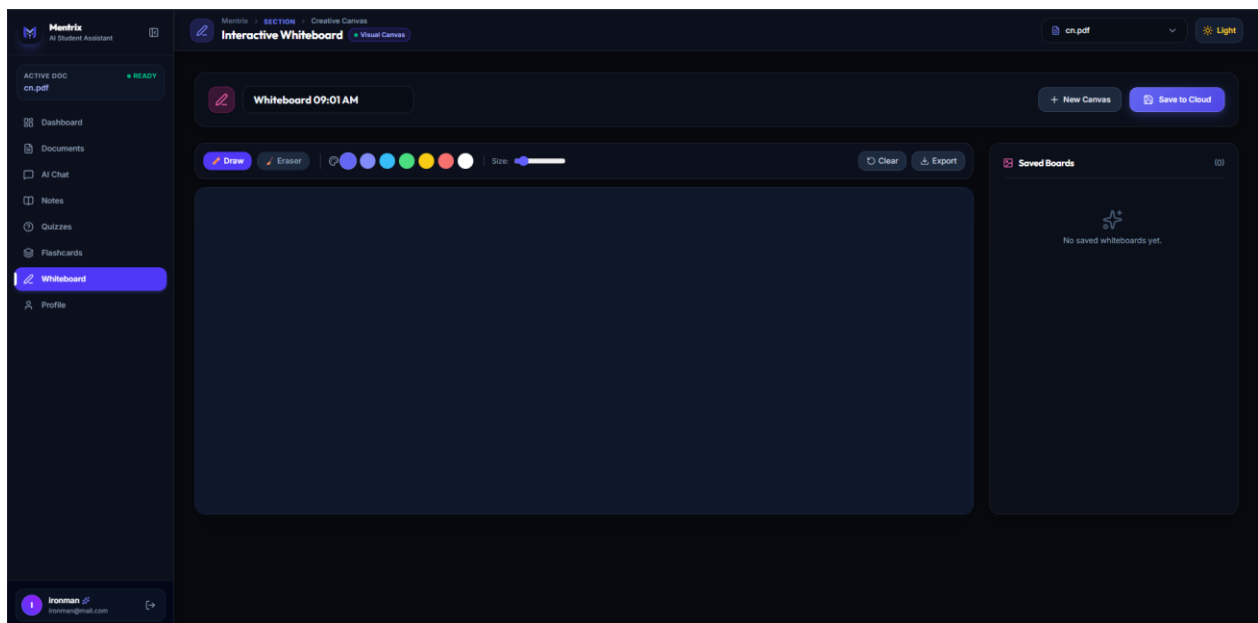


Figure 4.2.10: Whiteboard Interface

4.2.11 User and Academic Profile Interface

The Profile Interface allows students to manage their personal and academic information.

Main elements:

- Profile picture.
- Username and email.
- Account verification status.
- Gender and age information.
- College name.
- University name.
- Course/Major.

- Branch/Specialization.
- Year.
- Semester.
- Roll number.
- **Save Academic Details** button.

The screenshot displays the 'User & Academic Profile' interface in a dark-themed application. On the left is a sidebar with navigation options like Dashboard, Documents, AI Chat, Notes, Quizzes, Flashcards, and Whiteboard, with 'Profile' highlighted. The main content area shows the user's profile information and a form for academic details. The profile section includes a name 'ironman', email 'ironman@mail.com', gender 'MALE', and age '20'. The academic details form has fields for College Name, University Name, Course / Major, Branch / Specialization, Year, Semester, and Roll Number, with a 'Save Academic Details' button at the bottom right.

Figure 4.2.11: User and Academic Profile Interface

4.3 Deployment and Security Design

4.3.1 Deployment Design

Mentrix uses **Docker** and **Docker Compose** to provide a consistent environment for its different services.

Main Services

1. Frontend

- Runs the React.js web application.
- Provides the user interface.

2. Backend

- Runs the Node.js and Express.js application.
- Handles APIs and application logic.

3. AI Service

- Runs the Python FastAPI service.
- Handles AI and document-processing operations.

4. Background Worker

- a. Handles time-consuming document-processing tasks.

5. PostgreSQL

- a. Stores structured application data.

6. Redis

- a. Provides caching and queue support.

7. Qdrant

- a. Stores document embeddings.

8. MinIO

- a. Stores uploaded academic files.

Deployment Steps

1. Docker environment is prepared.
2. Required environment variables are configured.
3. Docker Compose starts the required services.
4. PostgreSQL and storage services are initialized.
5. Backend and AI services are started.
6. Background worker is started.
7. Frontend communicates with the backend through REST APIs.
8. All services work together to provide the Mentrrix application.

4.3.2 Security Design

Security mechanisms are implemented to protect student accounts, academic documents, and application data.

Authentication

- JWT is used for user authentication.
- Protected APIs require valid authentication.
- Unauthorized requests are rejected.

Password Security

- Passwords are hashed using bcrypt.
- Plain-text passwords are not stored.

User Data Protection

- Students can access their own academic resources.
- Protected routes verify the authenticated user.
- User-specific data is restricted through authorization.

File Validation

- Uploaded files are validated before processing.
- Unsupported file types are rejected.
- File size restrictions are applied.
- Invalid files are not processed.

Secret Management

- API keys and database credentials are stored using environment variables.
- Sensitive credentials are not directly included in source code.

4.4 Chapter Summary

The system design of Mentrix defines the technical workflow used to process academic documents and provide AI-powered learning assistance. The design includes document processing, OCR, embedding generation, RAG-based question answering, quiz generation, adaptive flashcards, and AI-generated notes. The system also provides dedicated interfaces for document management, learning activities, and digital note-taking. Docker-based deployment and security mechanisms such as JWT, bcrypt, API protection, and file validation provide a structured and secure environment for the application.